STATS 7022 - Data Science PG
Assignment 3

Trimester 1, 2025

Question 1: "Quick" Analysis

Sometimes the role of a data scientist is straightforward: clean and analyse a simple dataset. In this course, we have seen how to clean data; we have also learnt about a variety of complex models and how to fit them. In this question, we will clean and then analyse a dataset by fitting predictive models to it.

Question

You are to clean and analyse the `a3_data` dataset, available on MyUni. The data comprises observations of the following variables:

- `Y` - a categorical variable that takes values `A` or `B`.
- `X1` - a quantitative variable.
- `X2` - a quantitative variable.
- `X3` - a quantitative variable.
- `X4` - a quantitative variable.
- `C1` - a categorical variable that takes values `w`, `x`, `y`, or `z`.
- `C2` - a categorical variable that takes values `T`, `U`, or `V`.

Your analysis should comprise the following steps:

(a) Read the data into `R`.

(b) Produce EDA tables using skimr.

(c) Clean the data by:

  i. Ensuring that all variables are of the correct type.
  ii. Identifying and removing any outliers or unexpected values from the numeric variables.
  iii. Identifying and removing any outliers or unexpected values from the categorical variables.
  iv. Identifying and removing any observations with missing values (`NA`s).

(d) Create a tidymodels recipe with `Y` as the response, all other variables as predictors, and the following pre-processing steps:

   i. Normalise all numeric predictors.
   ii. Create binary indicator (dummy) variables from all categorical predictors.

(e) Create two support vector machine (SVM) models with the following kernels:
   i. Polynomial (degree 3).
   ii. Radial Basis Function (RBF).

(f) Create workflows for the two SVM models.

(g) Fit each of the models to the full dataset (you do **NOT** need to use a train/test split, cross-validation, or bootstrap validation).

(h) Use each fitted model to predict `Y` for the full dataset, and calculate the following performance metrics:
   i. Accuracy.
   ii. Sensitivity.
   iii. Specificity.

(i) Calculate and plot a ROC curve for each fitted model, and calculate AUC for each ROC curve.

(j) Choose the better model among the two SVMs.

(k) Use the better model to predict `Y` for the following new observation:

Predictor	**Value**
`X1`	-0.11
`X2`	1.19
`X3`	-1.37
`X4`	-0.46
`C1`	y
`C2`	T

As this is a "quick" analysis, you are not required to provide captions for any tables or plots you generate (but it is still a good idea to do so!).

For further information, see the assessment rubric on MyUni.

[Question total: 15 marks]

Submission

A single pdf file documenting your analysis of the `a3_data` dataset using support vector machines.

Checklist

Please ensure that:

- Your submission includes all `R` output and plots to support your answers where necessary.
- Your submission includes all `R` code.
- Your submission includes a caption for every plot and table.
- Your submission is a single pdf file - correctly oriented, clear, and legible.
- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

The course's regular late policy applies to this assignment. If you need to request an extension, or request a variation to the assessment on medical or compassionate grounds, please contact the course coordinator.

Question 2: Smoothing and Natural Splines

An important part of being a data scientist is understanding the mathematics that underpins the methods we use. The deepest understanding includes recognising the links between different methods. In this question, we will look at the relationship between smoothing splines and natural splines.

Question

Consider a random sample $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. Suppose that $g(x)$ is a natural cubic spline with knots $a < x_1 < \ldots < x_n < b$, i.e., $g(x_i) = y_i$. Let $\tilde{g}(x)$ be any other twice continuously differentiable function, such that $\tilde{g}(x_i) = y_i$, $i = 1, \ldots, n$.

(a) Let $h(x) = \tilde{g}(x) - g(x)$. Show that $\displaystyle\int_a^b g''(x)h''(x)dx$

$$= g''(x)h'(x)\Big|_a^{x_1} + \sum_{j=1}^{n-1}\left(g''(x)h'(x)\Big|_{x_j}^{x_{j+1}}\right) + g''(x)h'(x)\Big|_{x_n}^{b}$$
$$- \int_a^{x_1} g'''(x)h'(x)dx - \sum_{j=1}^{n-1}\left(\int_{x_j}^{x_{j+1}} g'''(x)h'(x)dx\right) - \int_{x_n}^{b} g'''(x)h'(x)dx.$$

[2 marks]

(b) Hence, show that

$$g''(x)h'(x)\Big|_a^{x_1} + \sum_{j=1}^{n-1}\left(g''(x)h'(x)\Big|_{x_j}^{x_{j+1}}\right) + g''(x)h'(x)\Big|_{x_n}^{b} = 0.$$

[3 marks]

(c) Hence, show that

$$\int_a^b g''(x)h''(x)dx = -\sum_{j=1}^{n-1}\left(\int_{x_j}^{x_{j+1}} g'''(x)h'(x)dx\right).$$

[2 marks]

(d) Hence, show that

$$\int_a^b g''(x)h''(x)dx = 0.$$

[2 marks]

(e) Hence, show that

$$\int_a^b \tilde{g}''(x)^2dx \geq \int_a^b g''(x)^2dx.$$

[4 marks]

(f) Hence, show that

$$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx$$

if and only if $h(x) = 0$ for all $a < x < b$.

[4 marks]

(g) If we are trying to find a smoothing spline

$$\hat{f}(x) = \min_f \left\{ \sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \int_a^b f''(x)^2 dx \right\}$$

show that this must be a natural cubic spline with knots at $x_1, \ldots, x_n$.

[4 marks]

[Question total: 21 marks]

Submission

A single pdf file containing your solutions. You may handwrite your answers and scan them, or typeset your answers.

Checklist

Please ensure that:

- Your submission is a single pdf file - correctly oriented, clear, and legible.
- You have shown all of your working, including probability notation where necessary.
- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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Question 3: k-Means

One of the best ways to understand an algorithm is to implement it yourself. In this question, we will write a function to implement the k-means algorithm.

Question

In `R`, write a function called `get_kmeans()`. Your function should take two inputs:

- `x`: a numeric vector.
- `k`: the number of groups.

Your function should return a vector of labels indicating group membership for each value in `x`. You should format your labels as follows:

- the labels of the groups should be the numbers $1, \ldots, k$.
- the groups should be ordered such that group 1 has the smallest mean, group 2 has the second smallest mean, and so on, up to group k has the largest mean.

Here is an example to illustrate its use:

```
x <- rep(2:4, each = 3)
get_kmeans(x = x, 3)
```

```
## [1] 1 1 1 2 2 2 3 3 3
```

Here is a template you can use:

```
# Function to implement k-means algorithm
#
# INPUT:
#   x: numeric vector
#   k: number of groups
#
# OUTPUT
# A vector of integer labels indicating group membership.

get_kmeans <- function(x, k) {
  # YOUR CODE HERE
}
```

Your function should also ensure that the correct type of input is passed to the function and that edge cases are handled appropriately. For further information, see the assessment rubric on MyUni.

[Question total: 7 marks]

Submission

A single `R` file containing your function. Your code will be automatically unit-tested, so any changes to names etc. may result in a loss of marks. Your function must not use k-means functions from other packages - the code must be your own work.

Checklist

Please ensure that:

- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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